higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

ENGINEERING PROCESSES
NQF LEVEL 4

19 NOVEMBER 2019

This marking guideline consists of 6 pages.

Copyright reserved

QUESTION 1

1.1
- Viscosity (density or thickness)
- Additives (chemicals added to lubricant)
- Flashpoint
- Pressure and heat they it can withstand
- Shelf life
- How well it is for the machine or part to be used on (Any 4 × 1) (4)

1.2 1.2.1 It is the lowest temperature at which a gas or a liquid will ignite when a flame is applied

1.2.2 It is the density or thickness of the lubricants (2 × 1) (2)
[6]

QUESTION 2

2.1
- Lighting
- Ventilation
- Dust
- Dampness
- Hazards (5)

2.2 High pressure spray pump (1)

2.3
- The records will help factory owners to decide on the type of maintenance to be done.
- The records will show when the maintenance should take place.
- The records should explain the types of spare parts that should be kept in stock or when they should be bought. (3 × 1) (3)

2.4
2.4.1 True
2.4.2 True
2.4.3 False
2.4.4 True
2.4.5 False
2.4.6 True
2.4.7 False
2.4.8 False
2.4.9 True
2.4.10 True

(10 × 1) (10)

2.5
- Structural defaults/damage
- Casting soundness
- Design errors
- Uneven wear
- Poor workmanship
- Unauthorised Modifications (Any 5 × 1) (5)

2.6 2.6.1 Casting is when molten metal is poured into a mould

2.6.2 Friction is the resistance that prevents one surface from sliding easily over another

(2 × 1) (2)
[26]

QUESTION 3

3.1
3.1.1 G
3.1.2 D
3.1.3 F
3.1.4 B
3.1.5 A
3.1.6 J
3.1.7 L
3.1.8 H
3.1.9 M
3.1.10 K

(10 × 1) (10)

3.2
- Soluble oil flow capacity
- Backlash of feed screw
- Vibration of unit
- Machine noise (4 × 1) (4)

3.3 By ensuring that special containers are provided for different types of waste products. (2)
[16]

QUESTION 4

4.1
- Copper
- Zinc (2)

4.2 The ability of metal to bear pressure or weight without breaking. (2)

4.3
- Mechanical cutting
- Gas cutting
- Plasma cutting
- Laser cutting (4)

4.4
- The size and weight of the material.
- The type of cutting that needs to be done.
- The quantity that needs to be cut and the time allowed for the cutting process.
- The type of surface finish required.
- The machine available in your workshop or factory. (5)

4.5
- Ensure that the cut is straight and smooth.
- Check that the component conforms to the shape on the drawing instructions.
- Check that the dimensions of the cut piece are correct.
- Check that the tolerances comply with those on the drawing information.

(4 × 2) (8)

[21]

QUESTION 5

5.1 It is computer software used to programme numerical control machines. (2)

5.2
5.2.1 CAM stands for computer-aided machine
5.2.2 APT stand for automatically programmed tool.

(2 × 1) (2)

5.3
5.3.1 F-codes
5.3.2 G-codes
5.3.3 N-codes
5.3.4 T-codes
5.3.5 S-codes

(5 × 1) (5)

5.4
- By showing what types of surface finishes and edges are required in your drawing
- Checking that the finishes and edges correspond with those specified on the design drawing. (2 × 2) (4)

5.5
- Check the graphics of the completed components and see if they satisfy the design requirements.
- If not, the program needs to be amended. (2 × 2) (4)

[17]

Copyright reserved

QUESTION 6

6.1
- The cost and availability of tools and equipment.
- The cost and availability of materials.
- The amount of labour involved and the cost of the labour.
- The time it takes to prepare for and carry out the process.
- The equipment needed to join the materials. (5)

6.2
- What its fatigue point is
- How ductile the material is
- Whether the tension of the joined material can be drawn into the wire
- How much it will deform if compressed
- The amount of pressure or weight it can tolerate before deformation takes place (5)

6.3. 6.3.1 Durability refers to how long something will remain in good condition even if it is used a lot.

6.3.2 Composition refers to the way different parts are made up of.

(2 × 2) (4)
[14]

TOTAL: 100

Copyright reserved